Temporal Modeling vs. Structured Preprocessing: A Controlled Comparison for Early Sepsis Prediction

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***Abstract*—Early prediction of sepsis in the intensive care unit (ICU) is a critical clinical challenge, where timely intervention can significantly reduce mortality. This paper investigates whether temporal sequence modeling (LSTM) or structured preprocessing (missingness-aware feature engineering) is the stronger lever for improving early sepsis prediction. Using the PhysioNet/CinC 2019 dataset of 40,336 ICU patients, we conduct a controlled 2×2 experiment across two preprocessing strategies (median imputation vs. forward-fill with missingness indicators) and two model families (XGBoost and LSTM). Results show that XGBoost with missingness-aware preprocessing (AUPRC = 0.1730, 95% CI [0.1636, 0.1825]) consistently outperforms LSTM across both strategies, and that preprocessing choice is the dominant factor. SHAP analysis reveals ICULOS, hospital admission time, and respiratory trends as the most predictive features. The best model generalises reasonably across two hospital systems (Set A AUPRC = 0.1824, Set B = 0.1625).**

***Index Terms*—sepsis prediction, ICU, XGBoost, LSTM, missingness indicators, SHAP, PhysioNet**

I. Introduction

Sepsis, defined as life-threatening organ dysfunction caused by dysregulated host response to infection [1], affects over 1.7 million adults annually in the United States and carries a mortality rate of 15–30%. Early detection remains difficult: sepsis progresses rapidly, its onset is diffuse, and clinical features overlap with other conditions. Automated early-warning systems that operate on routinely collected ICU data offer a promising path to earlier intervention.

Two broad approaches are common in the literature [2]. The first relies on temporal sequence models such as LSTMs [3] to capture complex dependencies across time. The second focuses on feature engineering, particularly on how missing values are handled, to extract more signal from sparse, irregularly sampled clinical measurements. In practice, ICU lab values can be missing for 60–80% of timesteps [4], making missing data treatment a first-order methodological choice.

This paper poses a simple question: *which matters more, temporal model architecture or preprocessing strategy?* We answer this through a controlled 2×2 experiment on the PhysioNet/CinC 2019 sepsis dataset [5], crossing two preprocessing strategies with two model families.

II. Dataset and Experimental Setup

A. PhysioNet/CinC 2019 Dataset

The dataset comprises 40,336 ICU patients from two hospital systems (Set A: 20,336; Set B: 20,000), each represented as an hourly time series of 40 clinical features: 8 vital signs, 26 laboratory values, and 6 demographic variables. The original `SepsisLabel` is a binary indicator shifted 6 hours before clinical onset to create an early-warning target (`EarlyLabel`). Patients with sepsis onset within the first 6 hours (706 patients) were excluded as no valid prediction window exists. The final cohort (39,630 patients, 5.62% sepsis prevalence) was split 70/15/15 into stratified patient-level train/validation/test sets, preserving class prevalence across all splits.

B. 2×2 Experimental Design

We define four conditions as the Cartesian product of two preprocessing strategies and two model families:

- **Strategy A**: Median imputation (per-feature training median) followed by standard scaling. Simple and widely used.
- **Strategy B**: Per-patient forward-fill of the last observed value, supplemented by binary missingness indicator columns for each of the 40 original features. This encodes *when* values were missing as an explicit signal [6], [7].

For XGBoost (Conditions A and B), 48 lag features were added per condition: 6 temporal statistics (lag-1, lag-2, lag-4, 4-hour rolling mean/std, 1-hour delta) for each of the 8 vital signs, applied after the patient-level split to prevent data leakage. Final feature counts: Condition A (88 features), Condition B (125 features). For LSTM (Conditions C and D), raw time series of 40 features (Strategy A) or 76 features (Strategy B) were fed as padded sequences up to 72 timesteps.

III. Methods

A. XGBoost

XGBoost [8] was trained with `scale_pos_weight = 43.3` to account for the 43:1 class imbalance. Hyperparameters were selected via grid search over $\{3, 4, 6\}$ tree depths and $\{0.05, 0.1\}$ learning rates, with 500 estimators and early stopping (patience 20) on validation AUPRC.

B. LSTM

A two-layer LSTM [3] with a per-timestep linear output head was implemented in PyTorch [9]. Packed padded sequences handled variable-length patient stays. Training used BCEWithLogitsLoss with `pos_weight = 43.3`, Adam optimiser, gradient clipping (`max_norm = 1.0`), and early stopping (patience 10). Grid search covered hidden sizes {64, 128}, dropout rates {0.2, 0.3}, and learning rates {0.001, 0.0005}.

C. Evaluation

The primary metric is AUPRC (Area Under the Precision-Recall Curve), which is robust to class imbalance and directly reflects the precision-recall trade-off relevant to clinical screening. AUC-ROC is reported as a secondary metric. Metrics were computed using scikit-learn [10]. Decision thresholds were selected by maximising F1 on the validation set. All reported metrics are accompanied by 95% bootstrap confidence intervals (1,000 iterations, patient-level resampling). SHAP TreeExplainer [11] was applied to both XGBoost models to assess feature importance.

IV. Results

A. Main Results

Table I presents the full 2×2 comparison. XGBoost with Strategy B (Condition B) achieves the highest test AUPRC of 0.1730 (95% CI [0.1636, 0.1825]), representing a 7.5× improvement over the random baseline (prevalence = 0.023). Both LSTM conditions perform substantially below their XGBoost counterparts, with Condition C (AUPRC = 0.0625) and Condition D (AUPRC = 0.0618) showing no meaningful benefit from the richer Strategy B preprocessing.

TABLE I
2×2 Experimental Results (Test Set)

Cond.	Model	Strategy	AUPRC (95% CI)	AUC-ROC
A	XGBoost	A	0.1672 [0.158, 0.176]	0.8325
B	XGBoost	B	**0.1730** [0.164, 0.183]	**0.8539**
C	LSTM	A	0.0625 [0.058, 0.068]	0.7599
D	LSTM	B	0.0618 [0.056, 0.068]	0.7597

Fig. 1 shows the AUPRC bar chart with confidence intervals and precision-recall curves for all four conditions. The non-overlapping confidence intervals between XGBoost and LSTM conditions confirm that the performance gap is statistically meaningful.

B. ROC and Calibration Analysis

Fig. 2 presents ROC curves and calibration plots. Both XGBoost conditions achieve AUC-ROC above 0.83, while LSTM conditions cluster around 0.76. Calibration analysis reveals that XGBoost models produce well-calibrated probability estimates (predicted risk scores track actual event rates closely), whereas LSTM models are poorly calibrated, producing compressed probability outputs near zero. This has direct clinical implications: XGBoost risk scores are more suitable for threshold-based alerting systems.

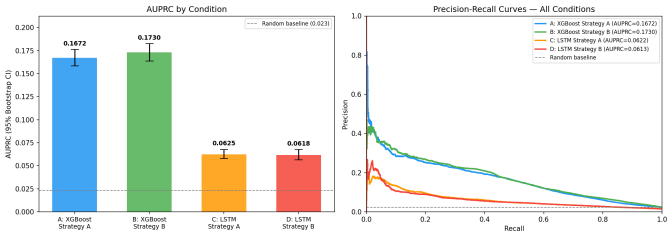

Fig. 1. AUPRC comparison with 95% bootstrap CIs (left) and precision-recall curves (right) for all four conditions. Dashed line indicates random baseline (prevalence = 0.023).

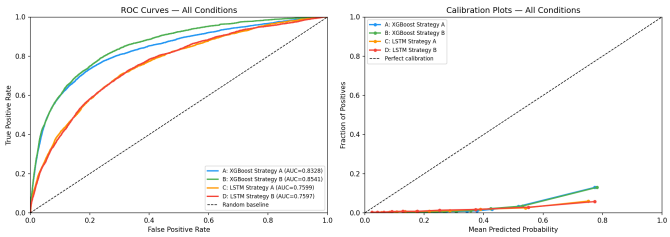

Fig. 2. ROC curves (left) and calibration plots (right). XGBoost models are better calibrated than LSTM models, making their predicted probabilities more reliable for clinical decision support.

C. Feature Importance via SHAP

Fig. 3 shows the top 20 features by mean absolute SHAP value for Conditions A and B. ICU length of stay (`ICULOS`) is the dominant predictor in both conditions, followed by hospital admission time and respiratory rate rolling mean, consistent with the clinical observation that respiratory deterioration tends to precede sepsis onset [12]. In Condition B, missingness indicators for FiO_2 and $EtCO_2$ appear in the top 20, consistent with the hypothesis that absent measurements carry predictive information. One plausible explanation is that certain tests are ordered selectively for deteriorating patients, making their absence a signal of relative stability [7].

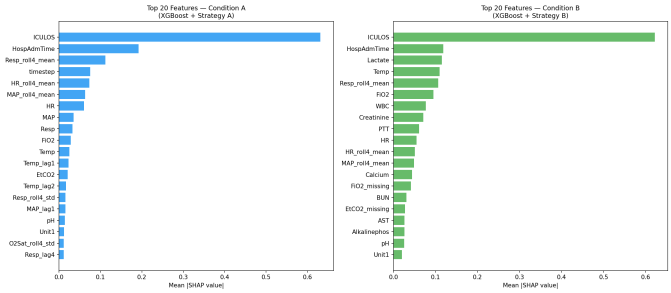

Fig. 3. Top 20 features by mean |SHAP value| for Condition A (left) and Condition B (right). Missingness indicators appear in Condition B, confirming their predictive value.

D. Hospital Generalizability

Fig. 4 shows XGBoost B performance stratified by hospital system. AUC-ROC is nearly identical (Set A: 0.8504, Set B: 0.8391), indicating that the model's discriminative ability generalises across institutions. AUPRC shows a modest gap (Set

A: 0.1824, Set B: 0.1625), likely attributable to differences in lab measurement practices and patient population between hospital systems.

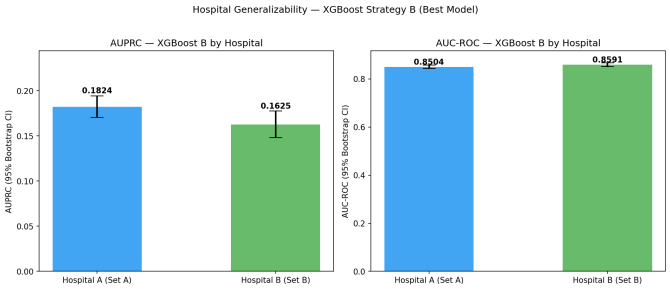

Fig. 4. XGBoost B performance on Hospital Set A vs. Set B test patients. AUC-ROC generalises well; AUPRC shows modest site-level variation.

V. Discussion

A. Why XGBoost Outperforms LSTM

The consistent superiority of XGBoost over LSTM on this dataset is explained by three dataset characteristics. First, ICU lab values are missing for 60–80% of timesteps: the sequences seen by the LSTM are largely imputed or forward-filled values rather than true observations. Second, the 43:1 class imbalance means that positive timesteps are heavily diluted within each sequence, making it difficult for the LSTM to identify the relevant signal window. Third, XGBoost's per-timestep treatment of the data, combined with explicit lag features, effectively captures temporal dynamics without requiring the model to learn sequential dependencies from sparse sequences. These findings align with prior work showing gradient boosting outperforms sequence models on irregular clinical tabular data [4].

B. The Role of Preprocessing

Strategy B improves XGBoost (AUPRC: 0.1672 $\rightarrow$ 0.1730); the confidence intervals partially overlap (A: [0.1582, 0.1761], B: [0.1636, 0.1825]), so this difference should be interpreted cautiously. By contrast, the gap between XGBoost and LSTM conditions is clearly non-overlapping and unambiguous. Strategy B has no meaningful effect on LSTM performance (0.0625 $\rightarrow$ 0.0618). This asymmetry suggests that the benefit of missingness-aware preprocessing is model-dependent: XGBoost can exploit missingness indicators as discrete features, while the LSTM's sequence-level optimisation cannot effectively leverage the same information across largely-missing sequences.

VI. Limitations and Conclusion

A. Limitations

This study has several limitations. The LSTM architecture used is a standard many-to-many design; architectures specifically designed for irregular time series, such as GRU-D [7], may close the performance gap by learning feature-specific decay rates for missing values. The lag feature engineering for XGBoost was limited to 8 vital signs; extending to lab values could further improve performance. Finally, while hospital generalizability is assessed at the test-set level, prospective validation across additional institutions remains necessary before clinical deployment.

B. Conclusion

We conducted a controlled 2×2 experiment comparing preprocessing strategy and model architecture for early sepsis prediction. XGBoost with missingness-aware preprocessing achieves the best performance (AUPRC = 0.1730), outperforming LSTM across both strategies. Preprocessing strategy is the dominant factor for XGBoost, while LSTM performance is largely insensitive to preprocessing, suggesting that the limiting factor for sequence models on this dataset is not feature representation but model-data alignment. Future work should explore architectures designed for irregular clinical time series and multi-site prospective validation.

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